

SET 1

Experiment 1: Smart Home Light Control System

Aim:

To design and simulate a Smart Home Light Control System using Arduino in Tinkercad for controlling household lighting using push-button based automation with brightness fading effects.

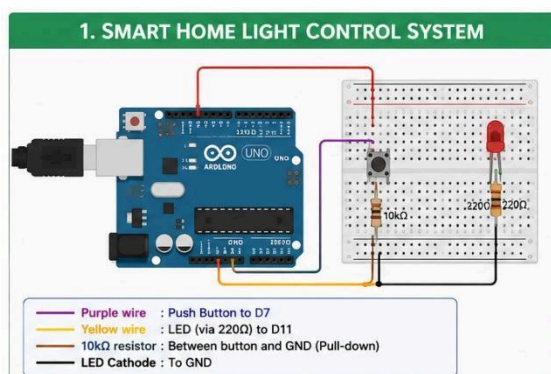
Hardware Required:

- Arduino UNO
- LED
- Push Button
- 220Ω Resistor
- 10KΩ Resistor
- Breadboard
- Jumper Wires
- USB Cable

Software Required:

- Tinkercad

Circuit Diagram Connections:



- Push Button → Digital Pin 7
- LED Positive Terminal → Digital Pin 11 through 220Ω resistor
- LED Negative Terminal → GND
- 10KΩ resistor used as pull-down resistor

Arduino Code:

```

#define BUTTON_PIN 7
#define LED_PIN 11

bool lightState = false;
bool lastButtonState = LOW;
unsigned long lastDebounceTime = 0;
unsigned long debounceDelay = 50;

int brightness = 0;
int fadeAmount = 5;

void setup()
{
    pinMode(BUTTON_PIN, INPUT);
    pinMode(LED_PIN, OUTPUT);

    Serial.begin(9600);
}

void loop()
{
    bool reading = digitalRead(BUTTON_PIN);

    if (reading != lastButtonState)
    {
        lastDebounceTime = millis();
    }

    if ((millis() - lastDebounceTime) > debounceDelay)
    {
        if (reading == HIGH)
        {
            lightState = !lightState;

            Serial.print("Light State Changed: ");
            Serial.println(lightState ? "ON" : "OFF");

            delay(300);
        }
    }

    lastButtonState = reading;

    if (lightState)

```

```

{
  analogWrite(LED_PIN, brightness);

  brightness += fadeAmount;

  if (brightness <= 0 || brightness >= 255)
  {
    fadeAmount = -fadeAmount;
  }

  delay(20);
}
else
{
  analogWrite(LED_PIN, 0);
}
}

```

Procedure:

1. Open Tinkercad Circuits and create a new circuit.
2. Add Arduino UNO, LED, push button, resistors, and breadboard.
3. Connect all components according to the circuit diagram.
4. Copy and paste the Arduino code into the code editor.
5. Start the simulation.
6. Press the push button and observe the LED brightness control.

Output:

- LED turns ON and OFF using push button.
- LED brightness continuously fades in and fades out.
- Serial Monitor displays the current light status.

Example:

```

Light State Changed: ON
Light State Changed: OFF

```

Result:

Thus, the Smart Home Light Control System with brightness fading effect was successfully designed and simulated using Arduino in Tinkercad.

Experiment 2: Temperature Monitoring System

Aim:

To develop and simulate an advanced Temperature Monitoring System using Arduino and LM35 temperature sensor in Tinkercad for real-time environmental monitoring.

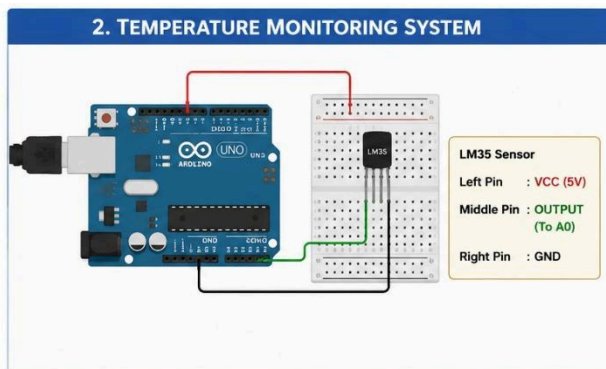
Hardware Required:

- Arduino UNO
- LM35 Temperature Sensor
- Breadboard
- Jumper Wires
- USB Cable

Software Required:

- Tinkercad

Circuit Diagram Connections:



- LM35 VCC → 5V
- LM35 Output → Analog Pin A0
- LM35 GND → GND

Arduino Code:

```
const int sensorPin = A0;  
  
float temperature;
```

```
float maxTemp = 0;
float minTemp = 100;

void setup()
{
    Serial.begin(9600);

    Serial.println("Temperature Monitoring System");
}

void loop()
{
    int sensorValue = analogRead(sensorPin);

    float voltage = sensorValue * (5.0 / 1023.0);

    temperature = voltage * 100;

    if (temperature > maxTemp)
    {
        maxTemp = temperature;
    }

    if (temperature < minTemp)
    {
        minTemp = temperature;
    }

    Serial.print("Current Temp: ");
    Serial.print(temperature);
    Serial.println(" C");

    Serial.print("Maximum Temp: ");
    Serial.print(maxTemp);
    Serial.println(" C");

    Serial.print("Minimum Temp: ");
    Serial.print(minTemp);
    Serial.println(" C");

    if (temperature > 35)
    {
        Serial.println("Warning: High Temperature
Detected");
    }
}
```

```

    }
    else if (temperature < 15)
    {
        Serial.println("Warning: Low Temperature
Detected");
    }
    else
    {
        Serial.println("Temperature Normal");
    }

    Serial.println("-----");

    delay(2000);
}

```

Procedure:

1. Open Tinkercad and create a new circuit.
2. Add Arduino UNO and LM35 sensor.
3. Connect the sensor pins properly.
4. Upload the Arduino code.
5. Start the simulation.
6. Open Serial Monitor to observe temperature readings and warnings.

Output:

- Real-time temperature values displayed on Serial Monitor.
- Maximum and minimum temperature tracking.
- Warning messages for abnormal temperatures.

Example:

```

Current Temp: 32.5 C
Maximum Temp: 35.0 C
Minimum Temp: 28.0 C
Temperature Normal

```

Result:

Thus, the advanced Temperature Monitoring System was successfully simulated using Arduino and LM35 sensor in Tinkercad.

Experiment 3: Distance Alert System

Aim

To design and implement an advanced Distance Alert System using ultrasonic sensor and Arduino in Tinkercad for obstacle detection and multi-level alert generation.

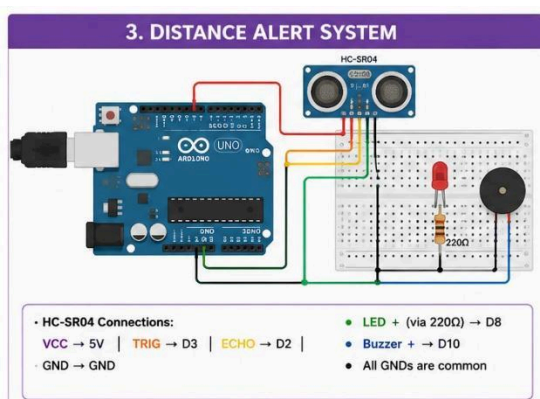
Hardware Required:

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Buzzer
- LED
- 220Ω Resistor
- Breadboard
- Jumper Wires
- USB Cable

Software Required:

- Tinkercad

Circuit Diagram Connections:



- Trigger Pin → Digital Pin 3
- Echo Pin → Digital Pin 2
- LED → Digital Pin 8
- Buzzer → Digital Pin 10
- Common GND connection

Arduino Code:

```
#define TRIG_PIN 3
#define ECHO_PIN 2
#define BUZZER_PIN 10
#define LED_PIN 8

long duration;
float distance;

void setup()
{
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    pinMode(BUZZER_PIN, OUTPUT);
    pinMode(LED_PIN, OUTPUT);

    Serial.begin(9600);

    Serial.println("Distance Alert System");
}

void loop()
{
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);

    digitalWrite(TRIG_PIN, LOW);

    duration = pulseIn(ECHO_PIN, HIGH);

    distance = duration * 0.034 / 2;

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    if (distance <= 10)
    {
        digitalWrite(LED_PIN, HIGH);
    }
}
```



```

    tone(BUZZER_PIN, 2000);

    Serial.println("CRITICAL ALERT");
}
else if (distance > 10 && distance <= 25)
{
    digitalWrite(LED_PIN, HIGH);

    tone(BUZZER_PIN, 1000);

    Serial.println("WARNING ALERT");
}
else
{
    digitalWrite(LED_PIN, LOW);

    noTone(BUZZER_PIN);

    Serial.println("SAFE ZONE");
}

Serial.println("-----");

delay(500);
}

```

Procedure:

1. Open Tinkercad and create a new circuit.
2. Add Arduino UNO, ultrasonic sensor, LED, and buzzer.
3. Make all required connections.
4. Upload the Arduino code.
5. Start the simulation.
6. Move objects near the ultrasonic sensor and observe the alert levels.

Output:

- LED and buzzer activate for nearby obstacles.
- Different buzzer frequencies generated based on object distance.
- Serial Monitor displays distance and alert status.

Example:

Distance: 8 cm
CRITICAL ALERT

Result:

Thus, the advanced Distance Alert System was successfully implemented using Arduino and ultrasonic sensor in Tinkercad.

Experiment 4: Home Automation System

Aim:

To create and simulate an advanced Home Automation System using Arduino in Tinkercad for intelligent control of household appliances.

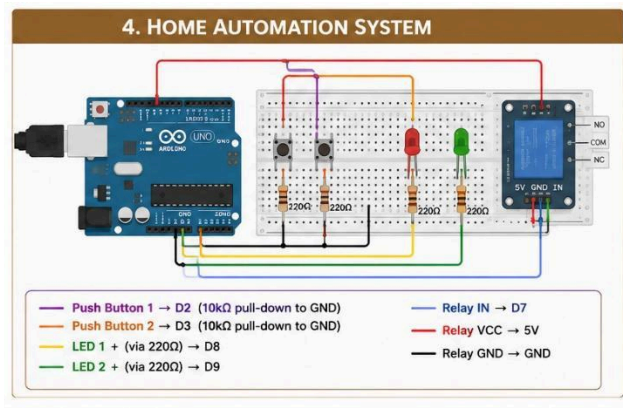
Hardware Required:

- Arduino UNO
- LEDs
- Relay Module
- Push Buttons
- 220Ω Resistors
- Breadboard
- Jumper Wires
- USB Cable

Software Required:

- Tinkercad

Circuit Diagram Connections:



- LED1 → Pin 8
- LED2 → Pin 9
- Relay Module → Pin 7
- Push Button1 → Pin 2
- Push Button2 → Pin 3

Arduino Code:

```
#define LIGHT1 8
#define LIGHT2 9
#define RELAY 7

#define SWITCH1 2
#define SWITCH2 3

bool state1 = false;
bool state2 = false;

bool prevSwitch1 = LOW;
bool prevSwitch2 = LOW;

void setup()
{
  pinMode(LIGHT1, OUTPUT);
  pinMode(LIGHT2, OUTPUT);
  pinMode(RELAY, OUTPUT);

  pinMode(SWITCH1, INPUT);
  pinMode(SWITCH2, INPUT);

  Serial.begin(9600);

  Serial.println("Smart Home Automation System");
```

```

}

void loop()
{
    bool currentSwitch1 = digitalRead(SWITCH1);
    bool currentSwitch2 = digitalRead(SWITCH2);

    if (currentSwitch1 == HIGH && prevSwitch1 == LOW)
    {
        state1 = !state1;

        digitalWrite(LIGHT1, state1);

        Serial.print("Light 1 Status: ");
        Serial.println(state1 ? "ON" : "OFF");

        delay(200);
    }

    if (currentSwitch2 == HIGH && prevSwitch2 == LOW)
    {
        state2 = !state2;

        digitalWrite(LIGHT2, state2);

        Serial.print("Light 2 Status: ");
        Serial.println(state2 ? "ON" : "OFF");

        delay(200);
    }

    prevSwitch1 = currentSwitch1;
    prevSwitch2 = currentSwitch2;

    if (state1 || state2)
    {
        digitalWrite(RELAY, HIGH);

        Serial.println("Home Appliances ACTIVE");
    }
    else
    {
        digitalWrite(RELAY, LOW);
    }
}

```

```
        Serial.println("All Appliances OFF");  
    }  
  
    delay(300);  
}
```

Procedure:

1. Open Tinkercad Circuits.
2. Add Arduino UNO, LEDs, relay module, push buttons, and resistors.
3. Connect all components according to the circuit diagram.
4. Upload the Arduino code.
5. Start the simulation.
6. Press the push buttons to control different appliances.

Output:

- Appliances turn ON/OFF through push button automation.
- Relay activates when any appliance is active.
- Serial Monitor displays appliance status.

Example:

```
Light 1 Status: ON  
Home Appliances ACTIVE
```

Result:

Thus, the advanced Home Automation System was successfully designed and simulated using Arduino in Tinkercad.